Reg. No. | | | | | | | | | | | [18MTAVS232]

SECOND SEMESTER M.TECH EXAMINATION, JUNE 2019

CONTROL SYSTEM ENGINEERING

Time: 03 Hours **Maximum Marks: 100**

Instructions:
- **i. Answer Five full questions.**
- **ii. Question numbers one and two are compulsory.**
- **ii. Draw neat sketches wherever necessary**

1. a. Describe **four** canonical form representation of LTI systems used in state variable analysis of control systems. **10**

b. Obtain a state-space model for the system shown in Figure 1(b) below: **10**

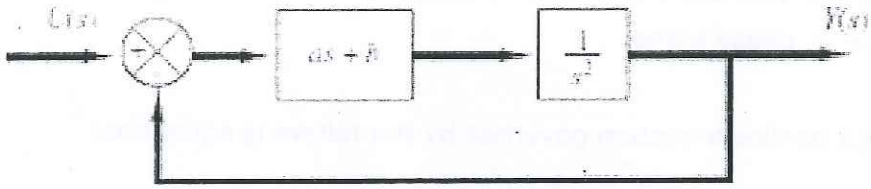

Fig 1(b)

2. a. Define state transition matrix (STM) and list its properties **05**

b. Define controllability and observability as applied to linear control system analysis? **05**

c. Obtain the transfer function of a LTI system given by the following state and output equations: **10**

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}-5 & -1\\ 3 & -1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}1\\ 1\end{bmatrix}u \quad \text{and } y = \begin{bmatrix}1 & 2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

3. a. What are the assumptions made and conditions required for assigning arbitrarily the location of poles in the design of linear control systems? **04**

b. A regulator system has a plant represented by the following transfer function: **16**

$$\frac{Y(s)}{U(s)} = \frac{10}{(s+1)(s+2)(s+3)}$$

Construct the state model defining suitable state variables.

Using state feedback control $u = -K^T x$, find K to locate the closed poles at $s_1 = -10$; $s_{2,3} = -2 \pm j\,2\sqrt{3}$

OR

4. a. What is state observer in the context of control system design? Discuss **04**

b. Consider the system defined by the state model given below: **16**

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 1.2 & 0.4 & -3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 1.2\end{bmatrix}u \text{ and } y = \begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

Design a full order state observer so that the observer will have the poles located at $s = -10$; $s = -5 \pm j\,5\sqrt{3}$.

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5. a. Discuss the behaviour of a nonlinear spring – mass – damper system with suitable sketches. **10**

b. What are singular points, what are the types of singular points encountered in phase plane analysis and how are they related to the Eigenvalues of the system under study? **10**

OR

6 a. Compare the behaviour of a nonlinear system with that of a linear system. **08**

b. For a linear system described by $\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}3 & -2\\ 2 & -2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$ identify the critical point. **12**

Compute and plot the Eigenvector trajectories corresponding to these Eigenvalues in the $x_1 - x_2$ phase plane.

7. Consider a nonlinear system governed by the following equations: **20**

$$\dot{x}_1 = -x_1 + 2x_1^2 x_2$$

$$\dot{x}_2 = -x_2$$

Analyse the stability of the system choosing a suitable Lyapunov function, and mark the stability constraints, if any, in the phase plane plot.

OR

8. a. State and discuss Lyapunov's stability theorems. **06**

b. Using the Lyapunov function $V(x) = x_1^2 + \mathbf{a}\, x_2^2$, where **a** is a positive constant, investigate the stability of the system described by the following equations: **14**

$$\dot{x}_1 = -x_1 + 4x_2$$

$$\dot{x}_2 = -x_1 - x_2^3$$